UCCD 2003 Object Oriented System Analysis and Design

Assignment (February 2025 Trimester)

1.0 Learning Outcome

- Model an Information System using the Unified Modeling Language (UML)

2.0 Instruction

i. Evidence of plagiarism or collusion will be taken seriously, and University regulations will be applied fully to such cases. You are advised to be familiar with the University's definitions of plagiarism and collusion. Any plagiarism or collusion may result in disciplinary action, in addition to **ZERO** mark being awarded to all parties involved.

ii. The assessment covers the topics from week 1 to week 8.

This assessment consists of 20 Individual points (I) and 60 Group points (G).

The total marks for assignment contribute 20% to the total grade.

The total marks for assignment calculated using the formula below:

*Total marks = [(total Individual points (I)*2 + total Group points (G)*1) / 100] *20*

iii. This is a **group assignment consists of 3-4 persons**.

iv. You are required to submit the softcopy of your assignment reports via WBLE links.

v. The deadline for the submission of the reports for **Task A** (softcopy) is on Monday, 17th March 2025, before 12 noon (Week 6).

The deadline for the submission of reports for **Task B** (softcopy) is on Monday, 28th April 2025, before 12 noon (Week 12).

Late submission will cause 10% marks being deducted per day.

3.0 Task to do.

Company Y plans to **build** their **membership card management system** in Malaysia. You are an employee of company Y and responsible to build the system. The stakeholders have given the user requirements as follows:

Customer perspectives:

1. Customers should be able to register for a membership card online. Associative
2. Customers must provide necessary personal information for registration.

Include (from 1)
(Validate
registration
information)

Associative

3. Customers should be able to update their profile information.
4. Customers must have the ability to apply for a membership card through the system. Associative

Include (from 4)

5. The application process should be user-friendly and include necessary documentation uploads.

Extend (from 4)

6. Once approved by staff, the system should generate and issue a unique membership card.

Associative

7. Customers should receive instructions on how to activate their membership card.

Include (from 6)

8. The system should provide automated reminders for membership renewal. Include (from 7)
9. Customers should be able to renew their membership online. Associative
10. Customers must have the ability to report a lost or stolen card through the system.

Associative

11. The system should facilitate the deactivation of lost or stolen cards. Include (from 10)
12. Customers should earn points for each purchase made using their membership card.

Associative

Include (from12) (update points)

13. The system should automatically update the customer's points balance after each eligible transaction.

Include (from 12)

14. Points should be awarded based on a predefined conversion rate (e.g., dollars spent per point).

Include (from13) (Calculate points)

15. Customers should have the ability to view their current points balance. Associative
16. The system should allow customers to redeem points for specified rewards or benefits.

Associative

Include (from 16) (rewards details)

17. Redemption options and corresponding point values should be clearly communicated to customers.

Include (from 15) (point expiry check)

18. The system should support the implementation of point expiry policies.
19. Customers should be notified in advance of points about to expire.

Extend (from18) (notify customer)

Extend (from 15) (deduct points)

20. Expired points should be deducted from the customer's balance.
21. Customers should be able to view a detailed transaction history, including points earned and redeemed.

Associative

22. The system should provide transparency regarding how points are earned and spent.
23. Implement a tiered membership system where customers can unlock additional benefits or earn points at an accelerated rate based on their membership level.

Associative (Unlock tier benefits)

24. Clearly communicate the requirements for achieving and maintaining each membership tier.

Associative (View membership tier requirements)

Staff perspectives:

1. Staff should have a dashboard to view and process incoming membership applications.

Associative (Manage membership application)

2. The system should support the efficient verification and approval of membership applications.

Include (from 1) (Verify membership application)

Associative

3. Staff should be able to issue new membership cards from an available inventory.

Incude (from 3) (Track membership card stock)

4. The system should track the stock and availability of physical membership cards.

Associative(view member information)

5. Staff should have access to a centralized database of member information.
6. The system should allow staff to update member profiles as needed.

Extend (update member profile)

Associative (manage membership renewals) (7)

7. Staff should be able to view and manage membership renewal requests.
8. Automated renewal reminders should be sent to members.

Include (from 7)

9. Staff should have a process to handle reported lost or stolen cards.

Associative

Include (from 9)

10. The system should deactivate lost or stolen cards to prevent unauthorized use.

Associative (view membership statistics) (11)

11. Staff should have access to reports on membership statistics, such as new registrations, renewals, and card issuances.
12. The system should provide analytics to help staff make informed decisions.

Include (from 11)

13. Staff should be able to send important notifications or updates to members through the system.

Associative

14. The system should log communication history with members.

Include (from13)

15. Staff should have access to a dashboard displaying an overview of point-related metrics, such as total points issued, redeemed, and outstanding.

Associative (view point-related metrics)

16. The system should support easy navigation and quick access to individual member point histories.

Associative

17. Staff should have the ability to manually adjust a member's points balance in case of errors or special circumstances.

Associative

18. Adjustments should be logged with a reason provided for auditing purposes.

Include (from 17)

19. Staff should be able to manage the catalog of rewards available for point redemption.

Associative (parent function)

20. The system should allow staff to add, modify, or remove rewards, including updating point values.

Associative (child functions)

21. Staff should have the ability to define and manage membership tiers.

Associative

22. The system should automatically adjust a member's tier based on predefined criteria.

Include (from 21)

23. Generate reports detailing point-related metrics for analysis and decision-making.

Associative (view reports)

24. Provide insights into popular rewards, point redemption trends, and member engagement.
25. Staff should be able to send targeted communications to members regarding their points balance, upcoming expirations, or special promotions related to point activities.

Associative

In addition, the project manager has prepared a workplan as follows:

Table 1: Workplan

No.	Task Descriptions	Duration	Task

		(Days)	Dependency
1	Kick off the project, set objectives, and identify key stakeholders.	5	-
2	Clearly define the scope of the project, including goals, deliverables, and constraints.	4	1
3	Schedule and conduct meetings with stakeholders to gather requirements and expectations.	5	2
4	Create a detailed project plan outlining tasks, timelines, and resource requirements.	3	2
5	Present the project plan to stakeholders for approval.	2	3,4
6	Define the high-level structure and components of the system.	5	5
7	Build an initial version of the system for early evaluation.	10	6
8	Gather feedback on the prototype and make necessary revisions.	7	7
9	Collect feedback from stakeholders on the revised prototype.	5	8
10	Incorporate feedback and finalize the detailed system design.	10	9
11	Implement the user interface of the system.	15	10
12	Build the backend logic and functionality.	5	10
13	Set up and integrate the database with the system.	4	12
14	Test the system for functionality, performance, and compatibility.	8	11,13
15	Address and fix any identified issues and optimize system performance.	10	14
16	Train end-users on how to use the system effectively.	12	15
17	Create comprehensive documentation for the system.	10	15
18	Conduct a final round of testing to ensure all issues are resolved.	5	16,17
19	Deploy the system to the production environment.	3	18
20	Close out the project, review outcomes, and obtain final approvals.	2	19

This project is handled by **3 to 4 team members**, which includes a group leader. The responsibility of the team members is to analyze and design a membership card management system. This project is divided into two major tasks as follows:

Task A (I – 5 points, G – 23 points)

i. **Draw the Gantt chart and network diagram.**

[Group work] Create a Gantt Chart and a network diagram illustrating the project timeline using software (see section 11.0) based on the information provided in Table 1. (G – 6 points)

ii. **Determine the critical path.**

[Group work] Compute the critical path using the information provided in Table 1. Show the calculation step by step including draw the network diagram, calculate early start, early finish, late finish and late start, slack time, and identify the critical path. (G – 11 points)

iii. **Mapping the tasks with rational unified process (RUP) workflows.**

[Group work] Arrange the tasks in the workplan to correspond with the workflows in the RUP, focusing on the related activities within each workflow rather than the project phases. (G – 5 points)

iv. **Review the existing system by examining the functionality**.

[Individual work] Each member will study and review the functionality as outlined below:

Member 1: The system shall allow customers utilize membership points for item(s) redemption.

Member 2: The system shall allow customers renew or reactivate their memberships.

Member 3: The system shall allow customers reporting lost or stolen cards and deactivating the cards.

Member 4: The system shall allow customers seek to return and refund item(s) redemption due to receiving incorrect or damaged items.

Find an existing system that is related to the functionality and discuss how it works. (I – 5 points)

v. Report format. (G – 1 point)

Task B (I – 15 points, G – 37 points)

i. **Membership card management system data flow diagram level 0.**

[Group work] Based on the user requirement given by stakeholders, draw a data flow diagram level 0 on a blank paper, and then take picture(s) of the drawing and include the pictures in the report. (G – 12 points)

ii. **System's functionalities for the membership card management system in use case diagram.**
[Group work] Based on the user requirement given by stakeholders, draw a use case diagram on a blank paper, and then take picture(s) of the drawing and include the pictures in the report. (G – 14 points)

iii. **Draw the DFD-0 and use case diagrams using software.**
[Group work] Based on the answers in the Task B part (i) and (ii), Draw the data flow diagram level 0 and the use case diagram using software (see section 11.0). (G – 6 points)

iv. **Fill up the use case descriptions and draw the activity diagrams.**
[Individual work] Each member in charge the following functionality:
Member 1: The system shall allow customers utilize membership points for item(s) redemption.
Member 2: The system shall allow customers renew or reactivate their memberships.
Member 3: The system shall allow customers reporting lost or stolen cards and deactivating the cards.
Member 4: The system shall allow customers seek to return and refund item(s) redemption due to receiving incorrect or damaged items.
Each member is required to fill up a use case descriptions template (refer to the sample in the main reference book, page 143, figure 4-13).
Then, draw an activity diagram based on the flow of events in the use case description on a blank paper, and then take picture(s) of the drawing and include the picture/s in the report. (I – 15 points)

v. **Membership card management system class diagrams.**
[Group work] Draw a class diagram using visual paradigm software (refer to the sample in the main reference book, page 177, figure 5-7). (G – 4 points)

vi. Report format. (G – 1 point)

4.0 Cover page sample

UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY

Session: February 2025

UCCD 2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

ASSIGNMENT:

Project title: Table Tennis Tournament Management System

Assignment group no.: 11

Member list:

No.	Group Member Name	Student ID	Programme	Signature
1				
2				
3				
4				

5.0 Sample Team Member Contribution in Group Tasks

Every member shall contribute to group works. The group marks will be deducted individually, if he/she has very less or no contribution in group works. Therefore, you are encouraged to distribute the group works fairly. In addition, you are required to reflect the contribution in the report too (See section 7.0).

5.1 Task A

Summarize the contribution of team members in Task A. For example:

**Group Task / Member No.	1	2	3	4
Determine the critical path	50%	50%	0%	0%
Draw the Gantt Chart	0%	0%	100%	0%
Mapping the tasks with rational unified process (RUP) workflows	0%	0%	0%	100%
Compile the report	100%	0%	0%	0%

Note: You may change the contribution weight (%). **Follow the sequence of member list in cover page.

5.2 Task B

Summarize the contribution of team members in Task B. For example:

**Group Task / Member No.	1	2	3	4
Membership card management system data flow diagram level 0	50%	50%	0%	0%
System's functionalities for the membership card management system in use case diagram	0%	0%	50%	50%
Draw the DFD-0 and Use case diagrams using software	0%	0%	50%	50%
Membership card management system class diagrams	50%	50%	0%	0%
Compile the report	0%	100%	0%	0%

Note: You may change the contribution weight (%). **Follow the sequence of member list in cover page.

6.0 Sample Table of Contents

6.1 Task A

6.2 Task B

2.4.4 Use case description [*4]

2.5 Activity diagrams

2.5.1 Activity diagrams [*1]

2.5.2 Activity diagrams [*2]

2.5.3 Activity diagrams [*3]

2.5.4 Activity diagrams [*4]

2.6 Class diagram diagrams

References

7.0 Report & Source Code Preparation

Each of the group members **MUST** contribute to this project. If anyone did NOT contribute to this project, he/she will be awarded ZERO mark to this group project. Any part contributed by the member or from any other sources should be given a credit / note / citation. Refer to the member list “No.” in the cover page (See section 4.0)

7.1 Report

Let’s say the whole section 2.1 was written by the 1st member, you can credit as below example:

2.1 Overview Functionality [*1]

Paragraph 1.

Paragraph 2.

Paragraph 3.

Let’s say the whole section 2.1 was written by **two members,** 1st and 2nd member, you can credit as below example:

2.1 Overview Functionality [*1, *2]

*Paragraph 1. [*1]*

*Paragraph 2. [*2]*

*Paragraph 3. [*1]*

It is better to avoid two members from writing one paragraph.

Let's say the sentences you refer from articles, websites, journals, books, other resources, or others peer works, you have to cite them using the **IEEE style citation**. The link for **IEEE style citation** documentation is in the IEEE website.

Let's say the source code you refer is from an existing program source code, other resources, or others peer works, you have to cite them using the **IEEE style citation**. The link for **IEEE style citation** documentation is in the IEEE website (https://library.utar.edu.my/documents/Guides/IEEEReferenceGuide_12Aug2022.pdf).

8.0 Manage image or picture size.

The image or picture in the report should be clear and easy to read. Every image or picture size should not be exceeded 500kb.

How to reduce the size of image or picture?

There are two ways to reduce the size of the image which is by using Microsoft Paint or Microsoft Word's "Compress Pictures" feature.

- **Using Microsoft Paint**
  - Open the image using Microsoft Paint (Window icon -> Window Accessories -> Paint)
  - Use "resize" and "rotate" option to remove unnecessary space.
  - Resize the width (horizontal) of image within the range of 300 pixels (~3 inch) to 750 pixels (~8 inch) and the height (vertical) of image within the range of 300 pixels (~3 inch) to 950 pixels (~10 inch).
  - Save the image into jpeg file format.

- **Microsoft Word "Compress Pictures" feature**
  - Place the image into the Microsoft Word.
  - Select the image, choose the "Format" tab.
  - Use "Crop" option (located at top right) to remove unnecessary space.
  - Resize the width (horizontal) of the image within the range of 300 pixels (~3 inch) to 750 pixels (~8 inch) and the height (vertical) of image within the range of 300 pixels (~3 inch) to 950 pixels (~10 inch).

- Select “Compress Pictures” option (located at top left) to compress the size of the image.

9.0 Report formatting

The report formats follow the final year project report submission guideline (https://fypfict.utar.edu.my/) stated in **Section 5**. The guideline in section 9.1 was referred from the faculty website with some minor changes.

9.1 Guideline

The report should be written by referring to a third person and in the past tense. For example, do not use "I" or "you" in the report.

- **Font**

－ Times New Roman, 12 points, 1.5 line spacing.

－ Applies to ALL, including figure caption, table caption, chapter headings and subheadings.

－ Exceptions:

Header, Footer, Footnote, Words in Figure/Table, font size should be within 10 to 11 points.

－ **Colour: black**.

－ Citing references in text: number the cross-references 1, 2, and so on, font size 12.

- **Language**

－ British English

- **Footer**

- Align right: page number.

- **Page Number**

- Align right at the Footer.

- Title, Abstract, Table of Contents and Listing – pages are numbered using small Roman numeric (a, ii, iii, etc). Note even though the Title Page is numbered it, the number is not to be printed on the page.

- Chapters and Bibliography – pages are numbered 1, 2, 3, etc.

- Appendices – pages are numbered A-1, A-2, etc for Appendix A, B-1, B-2, etc for Appendix B and etc.

Margins

－ Left (1 inches, except the front cover 1.2 inches)

－ Right (1 inch)

- Header/Footer (0.5/0.4 inch)

－ Top/Bottom (1 inches)

• **Title Page**

- Do not include UTAR logo.
- The font used is Times New Roman 12.
- Note the format (font type, size, capitalization and the sentences arrangement) of the

• **Table of Contents**

• **Tables/Figures**

－ Should include table (figure) caption immediately below the table (figure).

－ Number the tables and figures sequentially, with respect to the chapter or section of a chapter. To be consistent, use either one format, not both.

－ Make sure all the font in the figures is clear.

- For example, Table 2.2 is the second table of chapter 2.
- For example, Table 4.2.6 is the sixth table of section 2 of chapter 4

• **Citation**

－ Use IEEE style standard citation.

• **Bibliography**

－ Use IEEE style standard citation.

10.0 Report submission format

Submit the report in softcopy using the link at WBLE.

File name: Group-[group id]-Task [A or B]; e.g.: Group-01-Task A, Group-21-Task B.

File format: PDF or DOC or DOCX

Note: Every member MUST read the final compiled report and sign at cover page of the report.

11.0 Softwares

Gantt chart and network diagram tools:

1. Gantt Projects : https://www.ganttproject.biz/

2. ProjectLibre : https://www.projectlibre.com/product/1-alternative-microsoft-project-open-source

3. OpenProject community edition: https://www.openproject.org/community-edition/

4. Jira software basic (Gantt chart only): https://www.atlassian.com/software/jira/features/roadmaps?tab=basic

5. Microsoft Project

6. Other software

Data flow diagram tools:

1. Lucidchart Free plan: https://www.lucidchart.com/pages/examples/data-flow-diagram-software

2. Creately Free plan: https://creately.com/lp/data-flow-diagram-software-online/

3. Draw io : https://app.diagrams.net/

4. Microsoft Visio

5. Other software

UML diagrams tools

1. Visual paradigm community edition: https://www.visual-paradigm.com/editions/

2. Draw io : https://app.diagrams.net/

3. Other software